VOLCANOES AND VOLCANIC HAZARDS

A volcano is a rent or rupture in the earth's layer over which liquefied rock from under the earth's exterior erupts. According to Haddow, Bullock, and Coppola (2011), eventually, volcanoes will develop "upward and outward," developing mountains, islands, or large flat plateaus. Volcanic "mountains are formed through the accumulation of materials. such as lava, ash flows, and plates. (Haddow, Bullock, and Coppola, 2011, p.47).

A volcanic material that comes out from the earth is referred to as lava. The thinner lava moves quickly and becomes a large shield. The thicker lava forms steeper volcanic formations. If there is strong pressure from gases and molten rocks, these gases and rocks shoot up through the opening of the volcano and will cause explosions and violent eruptions (Haddow, Bullock, and Coppola, 2011).

TYPES OF VOLCANIC HAZARDS

1. **Volcanic Gases**. These gases are released into the atmosphere from the magma of the volcano during its eruption. These gases can also remain below the ground and rise toward the surface of the earth. In this case, gases may escape from the soil and volcanic vents into the atmosphere. The gases that are ejected by the volcanic eruption are harmful to those who are living around the volcano and to the atmosphere because these gases produce elements that create acid rains. Examples of these volcanic gases are hydrogen sulfide, carbon dioxide, hydrogen chloride, and hydrogen fluoride. Volcanic gases can rise and spread tens of kilometers into the atmosphere while explosive eruptions are happening. Winds may blow the clouds up to thousands of kilometers from a volcano once the gases are airborne. This will lead to the spread of gases from an erupting vent as tiny acid droplets, which are chemical compounds associated with the tephra particles.

2. **Lahar**. It is the flow of the hot/cold mixture of water and rock materials of a volcano going downstream through its river valleys and slopes. While lahar is moving, it looks like a mass of wet concrete that contains different rock debris from clay up to boulders. This volcano hazard may vary depending on its size and speed. Large lahars flow hundreds of meters wide and tens of meters in depth. They can flow several tens of meters per second, which is too fast for people to run out of the affected area. The speed and size of lahar changes as it goes downstream. Rock debris erodes and additional water can trigger a lahar to grow up to ten times their initial size. Lahar that flows down to river streams and plain fields can often cause environmental and economic damages. It can bury communities and valleys and block tributary water streams.

3. **Ash Fall**. It refers to the ashes ejected by a volcanic eruption, which consists of very small jagged pieces of glass and rock. These ashes are hard and mildly corrosive, and hard; conduct electricity when wet, and never dissolve in water. The ash may sometimes become so dense that the sunlight gets blocked. It spreads to broad areas by the wind. There is also a smell of sulfur that occurs during ash falls. The rock and glass particles attributed to this hazard can travel thousands of kilometers due to the wind, and they may result in injuries and damages to communities.

4. **Pyroclastic Flow**. It is a high-density mixture of dry and hot fragments of rocks and hot gases that move away from the volcano vent and move in high speed. This pyroclastic flow may result from a nonexplosive eruption of lava when thick lava flow collapses down a slope. It has two parts: the basal flow of fragments that is moving along the ground and a turbulent ash cloud that rises above the volcano's basal flow. This hazard destroys everything in its path. Containing rock fragments that vary in sizes, a pyroclastic flat can travel across slopes, and grounds can shatter or bury structures as well as objects out.

5. **Lava Flows**. These flow coming out from a volcano consist of molten rocks that ooze from an eruption. Lava is ejected during explosive and nonexplosive lava fountain activities and destroys every object in its path. However, people can have time to move away from it because of its slow movement. Lava has an extremely hot temperature that can knock over, bury, and surround the communities near a volcano.

In the Philippines, one of the most historic volcanic eruptions is the case of Mount Pinatubo in June 1991.

WHAT TO DO WHEN THERE IS A VOLCANIC ERUPTION

I. BEFORE

1. Stay alert and stay tuned-in to the radio and television for hazard updates.
2. Close all the doors and windows. This will prevent or reduce ashes from getting inside your house when there is an unexpected ash fall.
3. Bring your animals into closed areas/shelters.
4. Know the evacuation plan of your local community and listen to the authorities for instructions.
5. Keep your survival kit with you or within your reach.
6. Avoid staying in low areas that are prone to rockfalls and lava flows.
7. Prepare for possible evacuation.
8. Know the areas that are at risk from volcano hazards.
9. Know your local community emergency early warning signals.
10. Make and keep a list of emergency hotlines like those of the Philippine Red Cross and the National Disaster Risk Reduction Management Council.
11. If you are in a high-risk area, do a preemptive evacuation.

II. DURING

1. Stay awake and alert.
2. Listen to and follow the instructions given by the authorities in your area.
3. If caught by rockfall, protect your head and get away from the affected areas as soon as possible.
4. Cover your nose with a wet cloth to protect it from ashes falling on the area.
5. Wear long-sleeved shirts and pants.
6. Wear goggles and eyeglasses to protect your eyes from the ashes.
7. Never attempt to run a vehicle because this can stir-up volcanic ashes.
8. If trapped, call for rescue immediately.

III. AFTER

1. Clear the pathways and canals from ashes and debris.
2. Stay tuned-in to news updates.
3. Stay away from damaged power lines.
4. Check for injured and trapped persons without entering directly into the damaged area. Call for rescue if there are trapped persons.

LANDSLIDES

RAINFALL - INDUCED LANDSLIDES

Rainfall-induced landslide is a kind of hazard that occurs in many places in the world. It usually happens in mountainous places with a rainy environment. This kind of hazard is a result of prolonged rainfall that pushes the soil cover to collapse and causes a landslide. In some cases, the movement along the sliding surface of land leads to the crushing of soil grains, which causes liquefaction of the soil surface. Generally, landslides are triggered by rainfall, yet the incidence of a landslide is restricted basically by geological circumstances, morphological conditions, land cover conditions, and other related situations. The landslide susceptibility analysis can give a clue on the degree of danger site by site.

Landslides are dangerous because the earth slides down to the lower ground and contains a heavy mass of mud that can sweep off properties and bury people.

SINKHOLES

A sinkhole is also known as a "wallet snake hole" and is a natural hole on the Earth's surface. This hazard is caused by the chemical dissolution process in sandstone or the chemical dissolution of carbonate rocks that causes erosion of the soil beneath the ground. Sinkholes may form suddenly or gradually. They can also be a result of some human activities such as mining and over-extraction of subsurface fluids like groundwater.

Some sinkholes are also due to the substantial weight of material on top of the ground, which can trigger the underground collapse of bedrocks. Sinkholes are dangerous because these can take lives in an instant. They can occur without warning. People, houses, and establishments can be destroyed in a snap.

A sinkhole occurred in Florida on February 25. raising questions and concerns about this incredible phenomenon. The Florida Geological Survey (2005) claimed that sinkholes form in karst terrain principally from the breakdown of surface sediments into underground cavities in the limestone bedrock. Slightly acidic groundwater dissolves cavities and caves in the limestone over many years. The Earth falls if the cavity expands to the point that its ceiling can no longer support the weight of overlying sediments.

According to the United States Geological Survey (2013), about 20% of the US rests in areas vulnerable to sinkhole events.

WHAT IS A SINKHOLE?

Geologically speaking, a sinkhole is a depression in the ground that has no natural external surface drainage. Basically, this means that when it rains, all of the water stays inside the sinkhole and typically drains into the subsurface. Sinkholes are most common in what geologists call "karst terrain". These are regions where the type of rock below the land surface can naturally be dissolved by groundwater circulating through them. Soluble rocks include salt beds and domes, gypsum, limestone, and other carbonate rocks. Florida, for instance, is an area largely underlain by limestone and is highly susceptible to sinkholes.

When water from rainfall moves down through the soil, these types of rock begin to dissolve, and spaces and caverns develop underground. Sinkholes are dramatic because the land usually stays intact tor a period until the underground spaces just get too big. If there is not enough support for the land above the spaces, then a sudden collapse of the land surface can occur. Areas where water levels have lowered suddenly, are more prone to collapse formation.

DIFFERENT TYPES AND SEVERITIES

Sinkholes can be characterized into three types. Solution sinkholes are common in areas where there is very thin soil cover. Water erodes the area gradually, creating bowl-shaped depressions. Cover-collapse sinkholes develop abruptly (over a period of hours) and cause catastrophic damages. Cover subsidence sinkholes are formed over a long time. This usually happens when sand mostly covers the bedrock. The sand through openings in the bedrock, creating holes. Sinkholes can have dramatic effects, especially in urban settings. They can contaminate water resources and swallow up swimming pools, parts of roadways, and even buildings.

Important Things to Know about Sinkholes

1. It is a must for people to inspect the condition of their land properties and look for cracks in the foundation of a structure and small ground holes. Checking to see if living in areas underlain by soluble rocks can also be done by people by checking the geological surveys from county offices, local and state.
2. Sinkholes are indeed common in karst areas, but some sinkholes are formed because of human activity. Breakdowns may happen above old mines, and leaky faucets, destruction of sewerage, or groundwater pumping, and construction may also cause these.

3. If water-drainage patterns are altered and industrial and runoff-storage ponds are created, these can trigger an underground collapse.
4. Sinkholes may likewise occur due to aquifer systems. The sediments above the aquifer system may be balanced by ground-fluid water pressure to keep the surface soil in place. However, groundwater pumping for water supply and irrigation may produce sinkholes. The lowering of groundwater levels due to pumping may result in the failure of the underground system, which may result in sinkholes.
5. After heavy rains, sinkholes usually occur. However, it is important to note that droughts and freezes can also trigger the formation of sinkholes. As stated earlier, groundwater pressure helps support surface soil, but when droughts deplete underground water levels, the surface tends to weaken and consequently collapse when a saturating rain comes.

Where and When Sinkholes will Occur

Sinkholes are definitely dangerous as they result in property damage, contaminate the water supply, and kill people. Can we predict the formation of sinkholes?

It is difficult to predict exactly where and when sinkholes will occur. However, the Lake County Florida Department of Public Works provided clues that a sinkhole is forming. The clues are as follows:

- holes or concentric cracks in the ground
- sagging fence posts or trees
- newly exposed areas at the bottom of fence posts, trees, and
- foundations
- cracks in the foundation of a building
- other signs of shifting ground, such as a ceiling separating from the
- wall or improperly closing doors and windows
- cloudy water pumped from a previously clear source
- standing pools of rainwater in unusual places

WHAT TO DO WHEN THERE IS A VOLCANIC ERUPTION

I. BEFORE

1. Before purchasing a land or house, do research first if the area is prone to the occurrence of sinkholes.
2. Practice bringing a whistle every day as this can serve as your signaling device in case of being trapped in sinkholes.
3. If you are already living in an area where there is a frequent occurrence of sinkholes, consider leaving the area and moving to a safer place.

II. DURING

1. When you are trapped in a sinkhole, try to make noise but don't tap or dig the soil and rocks around you because there may be a possibility that your movement will create a soil erosion again.
2. If you have a cell phone and if you can still move, move slowly and carefully. Get your mobile phone and call for rescue.
3. Stay calm, no matter what happens. Save your body energy and wait for people that will pass by the sinkhole. Once it happens, call them for help.

III. AFTER

1. If you were caught in a sinkhole and you were able to escape it, check yourself for injuries and immediately seek help and report what happened to the authorities.
2. If you discover a sinkhole, report it immediately to the authorities.

REFERENCES:

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